

Yufan Du

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Education

Peking University *Sep. 2021 - Present*
Bachelor of Science in Applied Physics and Computer Science Dual Major *Beijing, China*

Scores

Overall GPA: 3.850/4.000 (3rd out of 64 students)
TOEFL: 108 (R27 + L30 + S23 + W28)

Skills

Proficient with: C/C++, CUDA, Python
Familiar with: Python ML libraries (PyTorch, NumPy), Verilog, FPGA, Embedded system, Linux system, Tcl, L^AT_EX, SQL/Database, JavaScript/HTML
EDA tools: Cadence Innovus, Cadence Virtuoso

Awards and Honors

ICCAD'24 Contest First Prize (1/45) (Team Leader)	ASU and NVIDIA	2024
Best Paper Nominee (5%)	ICCAD	2024
Industrial and Commercial Bank of China (ICBC)	ICBC	2024
Rising Star Scholarship (1%)		
Academic Innovation Award	Peking University	2024
Outstanding Student Leader Award (3%)	Peking University	2024
Huatai Securities Technology Scholarship	Huatai Securities	2023
Outstanding Student Leader Award (3%)	Peking University	2023
Huatai Securities Technology Scholarship	Huatai Securities	2022
Outstanding Student Award (10%)	Peking University	2022

Publications

- **Yufan Du**, Zizheng Guo, Yibo Lin, Runsheng Wang, and Ru Huang, “Fusion of Global Placement and Gate Sizing with Differentiable Optimization,” *2024 International Conference on Computer-Aided Design (ICCAD24)*. (**Best Paper Nominee**)
- Zizheng Guo, Zuodong Zhang, Wuxi Li, Tsung-Wei Huang, Xizhe Shi, **Yufan Du**, Yibo Lin, Runsheng Wang, and Ru Huang, “HeteroExcept: A CPU-GPU Heterogeneous Algorithm to Accelerate Exception-aware Static Timing Analysis,” *2024 International Conference on Computer-Aided Design (ICCAD24)*.
- **Yufan Du**[†], Zizheng Guo[†], Xun Jiang, Zhuomin Chai, Yuxiang Zhao, Yibo Lin, Runsheng Wang, and Ru Huang, “PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning,” *2024 Design Automation Conference (DAC24)*.
- Under review:
Yufan Du, Zizheng Guo, Yang Hsu, Zhili Xiong, Seunggeun Kim, David Z. Pan, Runsheng Wang, Ru Huang, and Yibo Lin, “Addressing Continuity and Expressivity Limitations in Differentiable Physical Optimization: A Case Study in Gate Sizing,” *2025 Design Automation Conference (DAC25)*.

Research Experiences

Differentiable NAS for Circuit Logic & Buffer Optimization Jun. 2024 - Present

- Developed a novel framework leveraging *neural architecture search (NAS)* for circuit logic optimization and buffer tree construction.
- Achieved superior PPA and reduced runtime compared to traditional methods.
- *Ready for submission to ICCAD 2025.*

Addressing Limitations in Differentiable Optimization Jun. 2024 - Sep. 2024

- Tackled *continuity* and *expressivity* limitations of differentiable optimization in real-world tasks.
- Proposed techniques, like *gradient clipping*, *gradient calibration*, and *progressive optimization*.
- Applied methods to a critical VLSI gate sizing task.
- *First Prize in ICCAD 2024 Gate Sizing Contest (1/45). & Submitted to DAC 2025.*

Global Placement and Gate Sizing Co-optimization Dec. 2023 - May. 2024

- Unified two originally separate VLSI design steps to *unlock larger optimization space*.
- Implemented gradient descent with differentiable PPA objectives by *CUDA programming*.
- Achieved 77.1% TNS and 43.5% WNS improvement with 1% lower power consumption.
- 7 $\times$ faster than traditional flow.
- *Accepted by ICCAD 2024 (Best Paper Nominee).*

Machine Learning for Chip Power Prediction Jun. 2023. - Nov. 2023

- Developed a framework for *early-stage chip power prediction* to guide optimization.
- Utilized a *GNN model* to predict cell and clock power, followed by a *CNN model* for refinement, forming a *ResNet structure*.
- Reduced total power prediction error from 9% to 2%.
- *Published by DAC 2024.*

Project Experiences

FPGA-Based Gesture-Controlled Snake Game Design Mar. 2023 - Jul. 2023

- Tackled the challenges of deploying algorithms efficiently on resource-limited devices.
- Implemented *computer vision algorithms*, including Canny edge detection, circular Hough transformation, and pattern recognition, alongside a game on a resource-limited FPGA.
- Enabled *real-time control* via a camera-based user gesture capture mechanism, offering valuable insights for *IoT edge device* applications.

AI-Assisted Schedule Manager Jan. 2023 - Apr. 2023

- Developed a *cross-platform application* integrated with natural language interaction capabilities for agenda scheduling.
- Enhanced user experience and helped them enjoy emerging *large language model (LLM)* benefits in real-world application.